

Problem 1.

Ask the user for the number of electricity units consumed in a month and print the bill. The first 100 units are free. The next 100 units (from 101 to 200) cost Rs. 2 per unit. Every unit above 200 costs Rs. 5 per unit.

(i) Before drawing anything, work out by hand what the program should print for each input below. (Once you have drawn your flowchart, trace it on these same inputs and check that it agrees with your answers here.)

Worked example: if units = 250, the first 100 units are free, the next 100 cost $100 \times 2 = \text{Rs. } 200$, and the remaining 50 cost $50 \times 5 = \text{Rs. } 250$, so the bill is Rs. 450.

input units	output
90	
150	
320	

(ii) Now draw a flowchart for the algorithm.

Problem 2.

Ask the user for a person's annual income and print the income tax. There is no tax on the first Rs. 2,50,000. Income from 2,50,001 to 5,00,000 is taxed at 5%. Income from 5,00,001 to 10,00,000 is taxed at 20%. Any income above 10,00,000 is taxed at 30%. (Only the part of the income that falls in each band is taxed at that band's rate.)

(i) Before drawing anything, work out by hand what the program should print for each input below. (Once you have drawn your flowchart, trace it on these same inputs and check that it agrees with your answers here.)

Worked example: if income = 6,00,000, the first 2,50,000 is tax-free, the next 2,50,000 is taxed at 5% = Rs. 12,500, and the remaining 1,00,000 is taxed at 20% = Rs. 20,000, so the tax is Rs. 32,500.

input income	output
200000	
400000	
1200000	

(ii) Now draw a flowchart for the algorithm.

Problem 3.

Ask the user for the total bill amount and print the amount payable after discount. There is no discount below Rs. 1,000. From 1,000 to 4,999 a 5% discount applies; from 5,000 to 9,999 a 10% discount; and 10,000 or above a 15% discount. (The discount applies to the whole amount.)

(i) Before drawing anything, work out by hand what the program should print for each input below. (Once you have drawn your flowchart, trace it on these same inputs and check that it agrees with your answers here.)

Worked example: if amount = 6,000, it falls in the 5,000–9,999 band, so a 10% discount applies and the amount payable is $6,000 \times 0.90 = \text{Rs. } 5,400$.

input amount	output
800	
3000	
15000	

(ii) Now draw a flowchart for the algorithm.

Problem 4.

Ask the user for a number n and print whether it is “Positive”, “Negative”, or “Zero”.

Draw a flowchart for this algorithm:

Problem 5.

A bank checks loan applications. Ask the user for the applicant's age and monthly salary. The applicant is "Eligible" only if their age is between 21 and 60 (both included) and their salary is at least Rs. 25,000; otherwise print "Not eligible".

Draw a flowchart for this algorithm:

Problem 6.

Ask the user for the lengths of the three sides of a triangle and print its type. If all three sides are equal it is "Equilateral"; if exactly two sides are equal it is "Isosceles"; if all sides are different it is "Scalene".

Draw a flowchart for this algorithm:

Problem 7.

Ask the user for a person's weight (in kg) and height (in metres) and print their BMI category. First compute $BMI = \text{weight} \div (\text{height} \times \text{height})$. A BMI below 18.5 is "Underweight", from 18.5 up to (but not including) 25 is "Normal", and 25 or above is "Overweight".

Draw a flowchart for this algorithm:

